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Catalytic Plastic Wastes Pyrolysis Fuel Upgrade/Separation and Its Application in Boiler

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Abstract Energy and environmental crises (especially plastic wastes pollution and GHG emissions) have jointly been spawning a new renewable alternative energy sources, derived from huge plastic wastes. In this work, pyrolysis oil was prepared from a 10 ton/day kiln pyrolysis facility with plastic wastes. During the pyrolysis, the mass and energy balances of plastic pyrolysis oil were understood from a commercial kiln pyrolysis facility. A 20-kg-scale vacuum distillation tower was utilized to separate the pyrolysis fuel into 78 fractions which were used to investigate the properties of pyrolysis fuel fractions at different AET (atmospheric equivalent temperature). A 2-kg-scale pyrolysis oil separator was used to have middle fraction pyrolysis oil (C11-C22), similar carbon ranges of petroleum diesel. The middle fraction pyrolysis oil from a 2-kg-scale separator represented a lower oxygen content due to the strong deoxygenation of catalysts. In addition, a uniform fuel mixture indicated a good miscibility of the separated pyrolysis oil and petroleum diesel. When the blends were used for combustion experiment in a 5 kW boiler, similar combustion and emission performance were observed. The boiler internal temperature was around 1,000-1,100°C when the efficiency was approximately 60.0%. Furthermore, the flue gas analysis indicated that the blends and diesel had similar concentrations of CO (about 10-30 ppm) and CO_2 (about 11-15 vol.%) depending on the excess air. However, a higher NOx concentration was observed with pyrolysis fuel blends (60-130 ppm) than that of pure petroleum diesel due to its higher nitrogen content in plastic waste pyrolysis fuel. Finally, this study suggested that pyrolysis is a very promising strategy for the utilization of plastic waste.

Keyword(s)
Plastic waste, Pyrolysis oil, Separation, Boiler combustion, Emission analysis

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